Akshat Porwal

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Education

Monash University Melbourne, Australia
Bachelor of Software Engineering and Bachelor of Commerce (Actuarial Studies) *Jan. 2023 – Expected Nov. 2027*

Technical Skills

Languages : Python, Java, JavaScript, TypeScript, SQL, R
Frontend : React, Next.js, Tailwind CSS, HTML, CSS
Backend and AI : FastAPI, REST APIs, asynchronous systems, RAG, vector search, embeddings, MCP, LLM evaluation
Data : PostgreSQL, Supabase, Redis, MongoDB, SQLAlchemy, indexing, query optimisation
Cloud and Infrastructure : AWS Lambda, Cognito, EC2, API Gateway, CloudFront, S3, SQS, ECR, AWS SAM, Vercel, Docker, Kubernetes, Terraform
Testing and Delivery : Pytest, Vitest, Playwright, accessibility testing, visual regression, GitHub Actions, CI/CD
ML and Analytics : Model evaluation, feature engineering, PCA, MCMC, dimensionality reduction

Experience

Technical Content Developer
DevOps & Cloud Labs

- Created technical slide decks covering DevOps and cloud engineering for a YouTube channel with more than 100,000 subscribers.
- Transformed complex technical concepts into structured, visually clear course material used in training products that generated more than $15,000 in sales.

Freelance Web Developer
247drain

- Designed and developed a responsive business website by gathering requirements directly from the client and iterating on stakeholder feedback.
- Integrated a retrieval-augmented generation (RAG) customer-support chatbot to answer service enquiries and help prospective customers find relevant information.
- Implemented website analytics and search engine optimisation practices to measure visitor behaviour, monitor conversions, and improve organic visibility.

Projects

LearnLens (QuantumLearn) | *Python, TypeScript, React, FastAPI, SQLAlchemy, Qiskit Aer, RAG* 2026

- Co-developed a role-based learning platform for introductory quantum computing, supporting educator course authoring, scaffolded student activities, simulation-grounded feedback, and administrative workflows.
- Implemented a course-scoped RAG pipeline for PDF, DOCX, and PPTX ingestion, structured extraction, heading-aware chunking, retrieval, and source-grounded feedback.
- Integrated retrieved material into task-generation and feedback workflows, adding evaluation, persistence, and recovery mechanisms for reliable processing.
- Designed versioned, privacy-safe evidence infrastructure with authorised capture adapters, append-only storage, learner-outcome snapshots, migrations, and automated tests.

Stonks in Hand | *Next.js, FastAPI, PostgreSQL, Supabase, AWS* 2026

- Converted the team's Figma concept into its initial MVP interface and improved production flows across announcements, search, ticker analysis, watchlists, and authentication.
- Co-designed the PostgreSQL and Alembic data model and FastAPI API foundation for watchlists, market data, reports, claims, scrape runs, and AI-analysis records.
- Built authentication across the frontend and backend, progressing from database-backed sessions to AWS Cognito with registration, MFA, password recovery, identity migration, and regression tests.
- Productionised staging with AWS SAM, Lambda, API Gateway, CloudFront, S3, SQS, ECR, CloudFormation, and GitHub Actions, resolving IAM, memory, concurrency, and rollback failures across multiple merged pull requests.

Codeology | *Next.js, React, TypeScript, FastAPI, Supabase, Vercel* 2026

- Extended and re-engineered an existing open-source AI-engineering platform, modernising its architecture, interface, and learning experience.
- Automated the generation and validation of 503 technical lessons across 20 structured learning phases.
- Designed hands-on pathways where users extend open-source projects, review AI-generated code for defects, and learn database and shell-scripting skills through interactive games.
- Built secure, account-backed AI CV analysis with bring-your-own-key provider support, document ingestion, vector retrieval, and personalised recommendations.
- Developed foundations for RAG chatbots and MCP-integrated learning assistants while establishing visual regression, accessibility, and automated CI/CD quality gates.

GRADIA | *Python, FastAPI, PostgreSQL, AWS RDS, LLM Pipelines* 2026

- Built an end-to-end AI content pipeline that transformed more than 1,000 uploaded PDF and DOCX files into structured lesson assets.
- Designed asynchronous processing workflows for documents exceeding 1,000 pages and PostgreSQL/RDS schemas for metadata, semantic tags, version histories, and generated outputs.
- Orchestrated multi-stage LLM pipelines that reduced manual lesson-creation time by 80%, average document-processing time by 35%, and rendering failures by 40%.

Fraud Detection System | *Python, Machine Learning, Model Evaluation*

- Built a fraud-detection pipeline for more than 250,000 transactions using feature engineering and five-fold cross-validation on a dataset containing only 3% fraudulent records.
- Improved Precision-Recall AUC from 0.81 to 0.92, reduced false positives by 28%, and lowered inference latency by 35% through model evaluation and feature selection.